

Unit 8: Special Equations and Classical Models

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. Problems 1–2 develop the Clairaut equation: general straight-line family and singular envelope elimination (Module 8.1). Problems 3–4 derive and solve the tractrix ODE and the general pursuit equation with a speed-ratio parameter (Module 8.2). Problem 5 solves the plane Poiseuille velocity profile with no-slip boundary conditions, and Problem 6 applies 2D incompressible continuity to find the stream function (Module 8.3). Problem 7 solves the linear price-adjustment ODE and identifies the equilibrium (Module 8.4). Problem 8 analyzes the full Lotka–Volterra predator-prey system including the conserved quantity H (Module 8.5). Problem 9 classifies coexistence versus competitive exclusion in a two-species competition model via nullcline inequalities (Module 8.6). Problem 10 closes with a four-part classification and solution survey (Module 8.7).

1. Find the general solution and the singular solution of the Clairaut equation $y = xy' + (y')^2$. Verify that the singular solution is the envelope of the general straight-line family.
2. For the Clairaut equation $y = xy' - \ln(y')$ (with $y' > 0$), find the one-parameter family of straight-line solutions and the singular solution. Confirm that the singular solution satisfies the ODE directly.
3. A target moves from the origin up the positive y -axis. A pursuer starts at $(a, 0)$ and is always connected to the target by a taut string of length a , with both moving at the same speed. Derive the tractrix differential equation

$$\frac{dy}{dx} = -\frac{\sqrt{a^2 - x^2}}{x}, \quad y(a) = 0,$$

from the constant-length tangent-line geometry, then solve it for $y(x)$ using the trigonometric substitution $x = a \sin \theta$.

4. A dog at the origin always runs directly toward its owner, who walks along the line $x = c$ in the positive y -direction at speed v_m . The dog runs at constant speed $v_d > v_m$. Let $k = v_m/v_d$. Derive the pursuit ODE $x y'' = k \sqrt{1 + (y')^2}$ with $y(c) = 0$, $y'(c) = 0$. Set $p = y'$, solve the resulting separable equation in p , then integrate to find $y(x)$ and the finite capture height $y(0)$.
5. A viscous fluid occupies the channel between two fixed plates at $y = 0$ and $y = h$. A constant pressure gradient $G > 0$ drives the flow; the fluid has dynamic viscosity μ and satisfies $\mu u''(y) = -G$ with $u(0) = u(h) = 0$. Solve this boundary-value problem for the velocity profile $u(y)$. State the location and value of the maximum velocity.
6. For a 2D incompressible steady flow the continuity equation requires $\partial u/\partial x + \partial v/\partial y = 0$. Given $u = x^2 - y^2$, find the most general $v(x, y)$ consistent with continuity. For the simplest choice $g(x) = 0$, determine the stream function $\psi(x, y)$ and write the equation of the streamlines.

7. Commodity price $p(t)$ obeys $dp/dt = k(D - S)$ with linear demand $D = a - bp$ and supply $S = -c + dp$ ($a, b, c, d, k > 0$). Write the ODE explicitly, solve it via integrating factor with $p(0) = p_0$, identify the equilibrium price p^* , and state the convergence condition.
8. For the Lotka–Volterra system $\dot{x} = ax - bxy$, $\dot{y} = -cy + dxy$ ($a, b, c, d > 0$): interpret each parameter ecologically; find all equilibria; show that the coexistence equilibrium is $(c/d, a/b)$; derive the conserved quantity $H(x, y) = dx - c \ln x + by - a \ln y$ by forming $dy/dx = \dot{y}/\dot{x}$ and integrating; verify that H attains its minimum at $(c/d, a/b)$.
9. Two competing species obey

$$\dot{x} = r_1 x \left(1 - \frac{x + \alpha y}{K_1} \right), \quad \dot{y} = r_2 y \left(1 - \frac{y + \beta x}{K_2} \right).$$

Find all four equilibria. Write the non-trivial nullcline equations. State the complete set of inequalities on K_1, K_2, α, β that determine (a) stable coexistence, (b) bistable competitive exclusion, (c) species x always wins, and (d) species y always wins. Derive the interior equilibrium (x^*, y^*) explicitly.

10. Classify each equation by the appropriate first-order solution method and carry out the complete solution.

(a) $\frac{dy}{dx} = \frac{x(1 + y^2)}{1 + x^2}$

(b) $\frac{dy}{dx} + \frac{2}{x}y = x^3$

(c) $\frac{dy}{dx} + y = xy^3$

(d) $(2xy + \cos x) dx + (x^2 - \sin y) dy = 0$

Part III — Complete Solutions

Solution 1. Set $p = y'$. The Clairaut equation is $y = xp + p^2$.

Differentiate both sides with respect to x ; since $dy/dx = p$:

$$p = p + x \frac{dp}{dx} + 2p \frac{dp}{dx} = p + (x + 2p) \frac{dp}{dx}.$$

Cancel p from both sides:

$$(x + 2p) \frac{dp}{dx} = 0.$$

Case A: $dp/dx = 0$. Then $p = C$ (constant). Substituting back: $y = Cx + C^2$. This is the **general solution**, a one-parameter family of straight lines.

Case B: $x + 2p = 0$. Then $p = -x/2$. Substitute into $y = xp + p^2$:

$$y = x \left(-\frac{x}{2} \right) + \left(-\frac{x}{2} \right)^2 = -\frac{x^2}{2} + \frac{x^2}{4} = -\frac{x^2}{4}.$$

Envelope verification via elimination. Write $F(x, y, C) = y - Cx - C^2 = 0$. Differentiate with respect to C : $\partial F/\partial C = -x - 2C = 0$, so $C = -x/2$. Substitute into F :

$$y - \left(-\frac{x}{2} \right) x - \left(\frac{x}{2} \right)^2 = y + \frac{x^2}{2} - \frac{x^2}{4} = y + \frac{x^2}{4} = 0 \implies y = -\frac{x^2}{4}.$$

Matches Case B. *Direct check:* $y' = -x/2$, so $(y')^2 = x^2/4$ and $xy' + (y')^2 = -x^2/2 + x^2/4 = -x^2/4 = y$. ✓

Solution 2. Set $p = y' > 0$. The equation is $y = xp - \ln p$.

Differentiate with respect to x :

$$p = p + x \frac{dp}{dx} - \frac{1}{p} \frac{dp}{dx} = p + \left(x - \frac{1}{p} \right) \frac{dp}{dx}.$$

Cancel p :

$$\left(x - \frac{1}{p} \right) \frac{dp}{dx} = 0.$$

Case A: $dp/dx = 0$. Then $p = C > 0$. **General solution:** $y = Cx - \ln C$ (family of lines).

Case B: $x - 1/p = 0$. Then $p = 1/x$. Substitute:

$$y = x \cdot \frac{1}{x} - \ln \left(\frac{1}{x} \right) = 1 + \ln x.$$

Singular solution: $y = 1 + \ln x$ ($x > 0$).

Envelope check. $F(x, y, C) = y - Cx + \ln C = 0$; $\partial F/\partial C = -x + 1/C = 0$, so $C = 1/x$. Substitute: $y - (1/x)x + \ln(1/x) = y - 1 - \ln x = 0$, giving $y = 1 + \ln x$. ✓

Direct check: $y' = 1/x$, $\ln y' = -\ln x$, so $xy' - \ln y' = 1 + \ln x = y$. ✓

Solution 3. *Geometric derivation.* At position (x, y) the pursuer's string of length a connects to the y -axis. The tangent to the path at (x, y) has slope dy/dx ; the horizontal distance to the y -axis is x , and by the right-triangle formed by the string, the horizontal leg equals x and the

hypotenuse equals a , so the vertical leg equals $\sqrt{a^2 - x^2}$. Since the pursuer is dragged *toward* the y -axis (decreasing x) and the string pulls downward in y relative to the target,

$$\frac{dy}{dx} = -\frac{\sqrt{a^2 - x^2}}{x}, \quad y(a) = 0.$$

Integration via $x = a \sin \theta$. Let $x = a \sin \theta$, $dx = a \cos \theta d\theta$, $\sqrt{a^2 - x^2} = a \cos \theta$:

$$\int \frac{\sqrt{a^2 - x^2}}{x} dx = \int \frac{a \cos \theta}{a \sin \theta} \cdot a \cos \theta d\theta = a \int \frac{\cos^2 \theta}{\sin \theta} d\theta.$$

Write $\cos^2 \theta = 1 - \sin^2 \theta$:

$$= a \int \frac{1 - \sin^2 \theta}{\sin \theta} d\theta = a \int (\csc \theta - \sin \theta) d\theta = a [\ln |\csc \theta - \cot \theta| + \cos \theta].$$

Back-substitute using $\csc \theta = a/x$, $\cot \theta = \sqrt{a^2 - x^2}/x$, $\cos \theta = \sqrt{a^2 - x^2}/a$:

$$= a \ln \left(\frac{a - \sqrt{a^2 - x^2}}{x} \right) + \sqrt{a^2 - x^2}.$$

Therefore:

$$y = -a \ln \left(\frac{a - \sqrt{a^2 - x^2}}{x} \right) - \sqrt{a^2 - x^2} + K.$$

Apply $y(a) = 0$: at $x = a$, $\sqrt{a^2 - x^2} = 0$ and $\ln(a/a) = \ln 1 = 0$, so $K = 0$.

Rationalizing the logarithm argument ($[a - \sqrt{a^2 - x^2}]/x = x/[a + \sqrt{a^2 - x^2}]$), the solution is

$$y(x) = a \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right) - \sqrt{a^2 - x^2}.$$

Solution 4. *Geometric setup.* At dog position (x, y) , the owner is at (c, Y) and the line from the dog to the owner is tangent to the path:

$$\frac{dy}{dx} = \frac{Y - y}{c - x} \implies Y - y = (c - x) y'. \quad (1)$$

Differentiate (1) with respect to x :

$$\frac{dY}{dx} - y' = -y' + (c - x) y'' \implies \frac{dY}{dx} = (c - x) y''. \quad (2)$$

The owner moves at speed v_m in y , the dog at speed v_d along its arc, so $dY/ds = k = v_m/v_d$ and $ds/dx = \sqrt{1 + (y')^2}$, giving:

$$\frac{dY}{dx} = k \sqrt{1 + (y')^2}. \quad (3)$$

Combining (2) and (3): $(c - x) y'' = k \sqrt{1 + (y')^2}$, which with origin shifted to the chased line reads

$$x y'' = k \sqrt{1 + (y')^2}.$$

Solving with $p = y'$.

$$x \frac{dp}{dx} = k\sqrt{1+p^2} \implies \frac{dp}{\sqrt{1+p^2}} = \frac{k dx}{x}.$$

Integrate:

$$\operatorname{arcsinh}(p) = k \ln x + C_1 \implies p + \sqrt{1+p^2} = A x^k.$$

Apply $y'(c) = p = 0$: $0 + 1 = A c^k$, so $A = c^{-k}$. Hence $p + \sqrt{1+p^2} = (x/c)^k$.

Since $(p + \sqrt{1+p^2})(\sqrt{1+p^2} - p) = 1$,

$$\sqrt{1+p^2} - p = (x/c)^{-k}.$$

Subtract from the first equation:

$$2p = (x/c)^k - (x/c)^{-k} \implies y' = \frac{1}{2} \left[\left(\frac{x}{c} \right)^k - \left(\frac{x}{c} \right)^{-k} \right].$$

Integrate ($k \neq 1$):

$$y = \frac{1}{2} \left[\frac{x^{k+1}}{(k+1)c^k} - \frac{c^k x^{1-k}}{1-k} \right] + C_2.$$

Apply $y(c) = 0$; at $x = c$ the bracket gives $c/(k+1) - c/(1-k)$, so:

$$C_2 = -\frac{1}{2} \left[\frac{c}{k+1} - \frac{c}{1-k} \right] = \frac{c}{2} \cdot \frac{2k}{1-k^2} = \frac{ck}{1-k^2}.$$

Solution:

$$y(x) = \frac{1}{2} \left[\frac{x^{k+1}}{(k+1)c^k} - \frac{c^k x^{1-k}}{1-k} \right] + \frac{ck}{1-k^2}.$$

Capture height (at $x \rightarrow 0$, since $k < 1$): $y(0) = ck/(1-k^2)$.

Solution 5. The governing BVP is $\mu u''(y) = -G$, $u(0) = 0$, $u(h) = 0$.

Integrate once:

$$\mu u'(y) = -Gy + C_1.$$

Integrate again:

$$\mu u(y) = -\frac{G}{2}y^2 + C_1y + C_2.$$

Apply $u(0) = 0$:

$$0 = 0 + 0 + C_2 \implies C_2 = 0.$$

Apply $u(h) = 0$:

$$0 = -\frac{G}{2}h^2 + C_1h \implies C_1 = \frac{Gh}{2}.$$

Substitute C_1 and C_2 :

$$u(y) = \frac{1}{\mu} \left(-\frac{G}{2}y^2 + \frac{Gh}{2}y \right) = \frac{G}{2\mu} y(h-y).$$

Maximum velocity: set $u'(y) = (G/2\mu)(h-2y) = 0$:

$$y_{\max} = \frac{h}{2}, \quad u_{\max} = u\left(\frac{h}{2}\right) = \frac{G}{2\mu} \cdot \frac{h}{2} \cdot \frac{h}{2} = \frac{Gh^2}{8\mu}.$$

Solution 6. Given $u = x^2 - y^2$, compute $\partial u / \partial x = 2x$. From continuity $\partial v / \partial y = -\partial u / \partial x = -2x$. Integrate in y :

$$v(x, y) = -2xy + g(x),$$

where $g(x)$ is an arbitrary function of x .

Stream function (with $g = 0$). Require $u = \partial \psi / \partial y = x^2 - y^2$. Integrate in y :

$$\psi(x, y) = x^2 y - \frac{y^3}{3} + h(x).$$

Check $-\partial \psi / \partial x = v = -2xy$:

$$-\frac{\partial \psi}{\partial x} = -(2xy + h'(x)) = -2xy \implies h'(x) = 0, \quad h = \text{const.}$$

Taking $h = 0$:

$$\psi(x, y) = x^2 y - \frac{y^3}{3}.$$

Streamlines are the level curves $\psi = k$:

$$x^2 y - \frac{y^3}{3} = k.$$

Verification that $dy/dx = v/u = -2xy/(x^2 - y^2)$: differentiate $\psi = \text{const}$ implicitly, $2xy + (x^2 - y^2)y' = 0$, giving $y' = -2xy/(x^2 - y^2) = v/u$. ✓

Solution 7. Substitute $D = a - bp$ and $S = -c + dp$:

$$\frac{dp}{dt} = k[(a - bp) - (-c + dp)] = k[(a + c) - (b + d)p].$$

Rearrange to standard linear form with $\beta = k(b + d)$ and $\alpha = k(a + c)$:

$$\frac{dp}{dt} + \beta p = \alpha.$$

Integrating factor $\mu(t) = e^{\beta t}$:

$$\frac{d}{dt}(p e^{\beta t}) = \alpha e^{\beta t}.$$

Integrate:

$$p e^{\beta t} = \frac{\alpha}{\beta} e^{\beta t} + C \implies p(t) = \frac{\alpha}{\beta} + C e^{-\beta t}.$$

Apply $p(0) = p_0$:

$$C = p_0 - \frac{\alpha}{\beta}.$$

Equilibrium price:

$$p^* = \frac{\alpha}{\beta} = \frac{k(a + c)}{k(b + d)} = \frac{a + c}{b + d}.$$

(Verify: $D = S$ gives $a - bp^* = -c + dp^*$, i.e. $a + c = (b + d)p^*$, same result.)

Solution:

$$p(t) = \frac{a + c}{b + d} + \left(p_0 - \frac{a + c}{b + d} \right) e^{-k(b+d)t}.$$

Convergence condition: $\beta = k(b + d) > 0$, which holds automatically for $k, b, d > 0$. Price converges monotonically to p^* for any p_0 .

Solution 8. *Parameter interpretation.* a = per-capita prey growth rate (in absence of predators); b = predation rate coefficient (prey removed per predator per unit time); c = per-capita predator death rate (in absence of prey); d = conversion efficiency (prey consumed into new predators).

Equilibria. Set $\dot{x} = 0$: $x(a - by) = 0 \Rightarrow x = 0$ or $y = a/b$. Set $\dot{y} = 0$: $y(-c + dx) = 0 \Rightarrow y = 0$ or $x = c/d$. Cross-pairing:

$$E_0 = (0, 0), \quad E^* = \left(\frac{c}{d}, \frac{a}{b}\right).$$

Derivation of H . Form $dy/dx = \dot{y}/\dot{x}$:

$$\frac{dy}{dx} = \frac{y(-c + dx)}{x(a - by)}.$$

Separate variables:

$$\frac{a - by}{y} dy = \frac{-c + dx}{x} dx \implies \left(\frac{a}{y} - b\right) dy = \left(-\frac{c}{x} + d\right) dx.$$

Integrate both sides:

$$a \ln y - by = -c \ln x + dx + \text{const.}$$

Rearrange to isolate the constant:

$$H(x, y) = dx - c \ln x + by - a \ln y = \text{const.}$$

H is minimized at E^* .

$$\frac{\partial H}{\partial x} = d - \frac{c}{x} = 0 \implies x = \frac{c}{d}, \quad \frac{\partial H}{\partial y} = b - \frac{a}{y} = 0 \implies y = \frac{a}{b}.$$

Since $\partial^2 H / \partial x^2 = c/x^2 > 0$ and $\partial^2 H / \partial y^2 = a/y^2 > 0$ with cross-partial zero, the Hessian is positive definite: E^* is the unique interior minimum. Level curves of H (closed) are the periodic phase-plane orbits around E^* .

Solution 9. *Equilibria.*

- $E_0 = (0, 0)$ (trivial extinction).
- $E_1 = (K_1, 0)$ ($y = 0$, $x = K_1$).
- $E_2 = (0, K_2)$ ($x = 0$, $y = K_2$).
- $E_3 = (x^*, y^*)$: solve $x + \alpha y = K_1$ and $\beta x + y = K_2$.

From $x + \alpha y = K_1$: $x = K_1 - \alpha y$. Substitute into $\beta x + y = K_2$:

$$\beta(K_1 - \alpha y) + y = K_2 \implies \beta K_1 + y(1 - \alpha\beta) = K_2 \implies y^* = \frac{K_2 - \beta K_1}{1 - \alpha\beta}.$$

$$x^* = \frac{K_1 - \alpha K_2}{1 - \alpha\beta}.$$

Non-trivial nullclines. x -nullcline: $x + \alpha y = K_1$, meeting axes at $(K_1, 0)$ and $(0, K_1/\alpha)$. y -nullcline: $\beta x + y = K_2$, meeting axes at $(K_2/\beta, 0)$ and $(0, K_2)$.

Outcome classification via axis intercepts.

(a) **Stable coexistence:** x -nullcline lies *above* the y -nullcline in the interior, i.e. both species inhibit themselves more than each other:

$$K_1 < \frac{K_2}{\beta} \quad \text{and} \quad K_2 < \frac{K_1}{\alpha} \quad (\Leftrightarrow \alpha\beta < 1).$$

Then $x^*, y^* > 0$ and E_3 is a stable node; E_1 and E_2 are saddles.

(b) **Bistable competitive exclusion:** the nullclines still cross in the positive quadrant but E_3 is an unstable saddle:

$$K_1 > \frac{K_2}{\beta} \quad \text{and} \quad K_2 > \frac{K_1}{\alpha} \quad (\Leftrightarrow \alpha\beta > 1).$$

Both E_1 and E_2 are stable nodes; initial conditions determine the winner.

(c) **Species x always wins:** $E_1 = (K_1, 0)$ globally stable:

$$K_1 > \frac{K_2}{\beta} \quad \text{and} \quad K_2 < \frac{K_1}{\alpha}.$$

(d) **Species y always wins:** $E_2 = (0, K_2)$ globally stable:

$$K_1 < \frac{K_2}{\beta} \quad \text{and} \quad K_2 > \frac{K_1}{\alpha}.$$

Solution 10.

(a) *Separable.* $dy/dx = x(1 + y^2)/(1 + x^2)$. Separate:

$$\frac{dy}{1 + y^2} = \frac{x dx}{1 + x^2}.$$

Integrate left: $\arctan y$. Integrate right via $u = 1 + x^2$, $du = 2x dx$:

$$\int \frac{x dx}{1 + x^2} = \frac{1}{2} \ln(1 + x^2).$$

Solution:

$$\arctan y = \frac{1}{2} \ln(1 + x^2) + C, \quad y = \tan\left(\frac{1}{2} \ln(1 + x^2) + C\right).$$

(b) *Linear first order.* $dy/dx + (2/x)y = x^3$. Integrating factor:

$$\mu = \exp\left(\int \frac{2}{x} dx\right) = e^{2 \ln x} = x^2.$$

Multiply:

$$x^2 y' + 2x y = x^5 \quad \Longrightarrow \quad \frac{d}{dx}(x^2 y) = x^5.$$

Integrate:

$$x^2 y = \frac{x^6}{6} + C \quad \Longrightarrow \quad y = \frac{x^4}{6} + \frac{C}{x^2}.$$

(c) *Bernoulli*, $n = 3$. $dy/dx + y = xy^3$. Divide by y^3 :

$$y^{-3} y' + y^{-2} = x.$$

Let $v = y^{-2}$; then $dv/dx = -2y^{-3}y'$, so $y^{-3}y' = -(1/2) dv/dx$:

$$-\frac{1}{2}v' + v = x \implies v' - 2v = -2x.$$

Linear in v . Integrating factor $\mu = e^{-2x}$:

$$\frac{d}{dx}(v e^{-2x}) = -2x e^{-2x}.$$

Compute $\int -2x e^{-2x} dx$ by parts with $u = x$, $dv = -2e^{-2x} dx$:

$$\int -2x e^{-2x} dx = x e^{-2x} - \int e^{-2x} dx = x e^{-2x} + \frac{e^{-2x}}{2}.$$

Integrate both sides:

$$v e^{-2x} = x e^{-2x} + \frac{e^{-2x}}{2} + C \implies v = x + \frac{1}{2} + C e^{2x}.$$

Back-substitute $v = 1/y^2$:

$$\frac{1}{y^2} = x + \frac{1}{2} + C e^{2x} \implies y^2 = \frac{1}{x + \frac{1}{2} + C e^{2x}}.$$

(d) *Exact.* $M = 2xy + \cos x$, $N = x^2 - \sin y$. Test: $\partial M/\partial y = 2x$ and $\partial N/\partial x = 2x$. Equal $\Rightarrow$ **exact**.

Integrate M with respect to x :

$$F(x, y) = \int (2xy + \cos x) dx = x^2 y + \sin x + g(y).$$

Differentiate F with respect to y and match to N :

$$F_y = x^2 + g'(y) = x^2 - \sin y \implies g'(y) = -\sin y \implies g(y) = \cos y.$$

General solution:

$$F(x, y) = x^2 y + \sin x + \cos y = C.$$

Unit 8 Key Takeaway

Every Clairaut equation $y = xy' + f(y')$ yields a **general line family** $y = Cx + f(C)$ and a **singular envelope** obtained by eliminating p from $x + f'(p) = 0$. Pursuit geometry converts constant-length or constant-speed constraints directly into separable or Bernoulli-type ODEs. The Poiseuille profile $u = (G/2\mu)y(h - y)$ and the price path $p(t) = p^* + (p_0 - p^*)e^{-k(b+d)t}$ both follow from first- or second-order linear BVPs. In Lotka–Volterra dynamics the **conserved quantity** $H = dx - c \ln x + by - a \ln y$ organizes all phase-plane orbits; in two-species competition the outcome is settled by $\alpha\beta < 1$ (**coexistence**) versus $\alpha\beta > 1$ (**exclusion**).